SINGAPORE DIMENSIONS • GRADE 6A • CHAPTER 1

Negative Numbers, Zero Pairs & Absolute Value

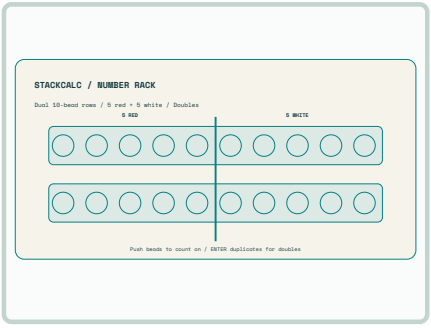
Tactile Concrete-Pictorial-Abstract (CPA) Station Guide • CCSS.MATH.6.NS.C.5

Student Name: _____ Date: _____ **Model: SC-6** **Check Score: ___ / 4**

STUDENT SELF-CHECK RULE (NO CHEATING):
Build with blocks and solve on paper first! Use your SC-6 calculator ONLY to verify answers on the 4-line LCD screen.

STATION LAB TUTORIAL: BENCH SCHEMATIC & CPA PROGRESSION

1. 2D Bench Blueprint



Kit: Number Rack

2. Desk Setup (Concrete -> Pictorial)

Physical Bench Setup:
Teal counters = +1; coral counters = -1.
Match 1 teal + 1 coral to form a zero pair.

Sketch Model on Paper:
Draw horizontal integer number line with zero benchmark.

3. Step-by-Step LCD Stack Trace (SC-6)

MODE:	123	Base-10 Arithmetic	
1.	8	Step 1	Execute [8]
2.	+/-	Step 2	Execute [+/-]
3.	ENTER	Step 3	Execute [ENTER]
4.	5	Step 4	Execute [5]

EXPECTED LCD X: -3

1 **Evaluate -4 + -6.**

Manipulative Model & Desk Sketch (Show your work):

Final Math Answer: _____

☐ Verified on SC-6

2 **Evaluate 7 - (-3).**

Manipulative Model & Desk Sketch (Show your work):

Final Math Answer: _____

☐ Verified on SC-6

3 **Evaluate -12 - (-5).**

Manipulative Model & Desk Sketch (Show your work):

Final Math Answer: _____

☐ Verified on SC-6

4 **Evaluate 15 + (-20).**

Manipulative Model & Desk Sketch (Show your work):

Final Math Answer: _____

☐ Verified on SC-6

Station Diagnostic & Solution Key: Chapter 1

Negative Numbers, Zero Pairs & Absolute Value • Manipulative Setup, Diagnostic Rubric & Verification Flow

MANIPULATIVE HARDWARE STATION INVENTORY & 3D PRINT PREPARATION

Station Kit: Ten-Frame Studio (Teal Positive, Coral Negative)

CAD Blueprint: number_rack

Target Device: SC-6 Elementary / SC-32 Bridge

CAD/ STL Source File:

Education/scad/number_rack_parts.scad

Standard: Gridfinity 42mm Pitch / 0.6mm Inset Markings

Recommended Filaments:

Teal (+1) and Coral (-1) Zero-Pair Beads (PLA)

#	Math Problem Prompt	Expected Math Answer	Self-Check Keystrokes & RPN Stack
1	**Evaluate -4 + -6.** Standard: CCSS.MATH.6.NS.C.5	-10	[4] [+/-] [ENTER] [6] [+/-] [+]
2	**Evaluate 7 - (-3).** Standard: CCSS.MATH.6.NS.C.5	10	[7] [ENTER] [3] [+/-] [-]
3	**Evaluate -12 - (-5).** Standard: CCSS.MATH.6.NS.C.5	-7	[1][2] [+/-] [ENTER] [5] [+/-] [-]
4	**Evaluate 15 + (-20).** Standard: CCSS.MATH.6.NS.C.5	-5	[1][5] [ENTER] [2][0] [+/-] [+]

TEACHER DIAGNOSTIC GUIDE: COMMON MISCONCEPTIONS & REMEDIATION

Key Misconception to Target During Station Observation:

Subtracting a negative means removing cold counters, which increases the temperature.

How the Manipulatives & Calculator Foster Self-Correction:

Students solve with pencil and physical blocks first. When they self-check on the SC-6, any discrepancy between their written answer and the calculator's 4-line LCD immediately prompts them to re-inspect their physical bench model or regrouping, turning errors into productive struggle without teacher intervention.

Station Facilitation & Pacing Tip:

Keep 3D printed trays docked on standard Gridfinity baseplates to prevent loose parts on student desks. Encourage students to articulate their CPA steps: name the block representation (Concrete), point to their drawn tape/bond model (Pictorial), and verify register values (Abstract).

CLASSROOM SCORING & STATION MASTERY RUBRIC

Exemplary (18–20 pts):	Complete, accurate manipulative sketches; exact math answers; independently verified self-check on LCD stack.
Proficient (14–17 pts):	Correct math answers; minor sketch omissions; verified on calculator with minimal teacher prompting.
Developing (10–13 pts):	Understands problem concept but made computational slips; successfully self-corrected using calculator LCD.
Intervention (<10 pts):	Relied on calculator before solving on paper, or struggled with underlying physical manipulative models.